

Tear fluid osmolarity as a potential marker of hydration status.



It has been suggested that tear fluid is isotonic with plasma, and plasma osmolality ($P(\text{osm})$) is an accepted, albeit invasive, hydration marker. Our aim was to determine whether tear fluid osmolarity ($T(\text{osm})$) assessed using a new, portable, noninvasive, rapid collection and measurement device tracks hydration. **PURPOSE:** This study aimed to compare changes in $T(\text{osm})$ and another widely used noninvasive marker, urine specific gravity (USG), with changes in $P(\text{osm})$ during hypertonic-hypovolemia. **METHODS:** In a randomized order, 14 healthy volunteers exercised in the heat on one occasion with fluid restriction (FR) until 1%, 2%, and 3% body mass loss (BML) and with overnight fluid restriction until 08:00 h the following day, and on another occasion with fluid intake (FI). Volunteers were rehydrated between 08:00 and 11:00 h. $T(\text{osm})$ was assessed using the TearLab osmolarity system. **RESULTS:** $P(\text{osm})$ and USG increased with progressive dehydration on FR ($P < 0.001$). $T(\text{osm})$ increased significantly on FR from 293 ± 9 to 305 ± 13 $\text{mOsm} \cdot \text{L}^{-1}$ at 3% BML and remained elevated overnight (304 ± 14 $\text{mOsm} \cdot \text{L}^{-1}$; $P < 0.001$). $P(\text{osm})$ and $T(\text{osm})$ decreased during exercise on FI and returned to preexercise values the following morning. Rehydration restored $P(\text{osm})$, USG, and $T(\text{osm})$ to within preexercise values. The mean correlation between $T(\text{osm})$ and $P(\text{osm})$ was $r = 0.93$ and that between USG and $P(\text{osm})$ was $r = 0.72$. **CONCLUSIONS:** $T(\text{osm})$ increased with dehydration and tracked alterations in $P(\text{osm})$ with comparable utility to USG. Measuring $T(\text{osm})$ using the TearLab osmolarity system may offer sports medicine practitioners, clinicians, and research investigators a practical and rapid hydration assessment technique.